

ADELAIDE MOUNT LOFTY, Australia

WMO: 956780

Lat: 34.9783S

Lon: 138.7089E

Elev: 685

StdP: 93.36

Time Zone: 9.50 (AUA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 2.5	(c) 3.1	(d) -9.4	(e) 1.8	(f) 18.3	(g) -5.9	(h) 2.5	(i) 17.8	(j) 17.8	(k) 7.9	(l) 15.7	(m) 7.3	(n) 4.9	(o) 110	(p) 0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 9.3	(c) 31.8	(d) 16.2	(e) 29.7	(f) 15.6	(g) 27.8	(h) 14.9	(i) 18.7	(j) 25.8	(k) 17.7	(l) 25.1	(m) 16.7	(n) 24.5	(o) 3.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.8	(b) 13.1	(c) 20.1	(d) 15.3	(e) 11.8	(f) 19.2	(g) 14.1	(h) 10.9	(i) 18.0	(j) 55.9	(k) 26.2	(l) 52.5	(m) 25.0	(n) 49.4	(o) 24.7	(p) 24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	12.5	10.7		1.0	34.5	0.5	2.2	0.6	36.1	0.3	37.4	0.0	38.7	-0.4	40.3
			DB	0.5	20.2	0.7	1.6	0.0	21.4	-0.4	22.3	-0.8	23.2	-1.3	24.3
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	17.5	17.6	15.5	13.0	10.0	7.6	6.9	7.4	9.3	11.5	13.7	15.6		
	DBStd	5.50	5.19	5.07	4.71	3.93	2.88	2.03	1.92	2.71	3.46	4.59	4.96	5.04		
	HDD10.0	419	1	1	2	11	34	76	98	91	54	33	13	4		
	HDD18.3	2477	82	72	119	166	258	321	354	338	271	219	158	119		
	CDD10.0	1190	233	213	173	102	36	5	3	12	34	81	123	177		
	CDD18.3	206	55	51	31	7	1	0	0	0	1	8	19	34		
	CDH23.3	2006	643	494	237	28	0	0	0	0	1	58	167	378		
	CDH26.7	659	255	163	52	4	0	0	0	0	0	12	41	132		
(14) Wind	WSAvg	4.5	4.2	4.0	3.9	4.1	4.4	5.0	5.4	5.2	5.2	4.7	4.1	4.0		
(15) Precipitation	PrecAvg	630	23	25	25	42	73	91	90	78	72	48	30	35		
	PrecMax	963	51	100	102	110	138	149	183	167	169	127	97	109		
	PrecMin	404	1	1	0	2	9	25	33	14	28	4	3	6		
	PrecStd	123	13	23	22	30	30	36	35	37	33	32	22	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	33.5	30.7	26.7	19.7	15.4	13.9	17.7	22.0	28.0	30.8	33.6		
		MCWB	17.0	18.3	15.5	14.6	11.1	10.0	8.4	10.5	12.0	14.3	15.8	16.7		
	2%	DB	32.1	30.9	28.0	23.4	17.7	13.1	12.2	15.4	19.8	24.7	27.4	30.4		
		MCWB	16.2	16.9	15.1	12.6	10.4	9.4	7.7	8.7	10.5	13.2	14.1	15.4		
	5%	DB	29.5	28.7	26.1	21.3	16.0	11.7	10.9	13.5	17.6	22.2	25.1	27.5		
		MCWB	15.5	15.7	14.7	11.8	10.1	8.8	7.3	8.1	9.8	12.1	13.5	14.1		
	10%	DB	27.0	26.5	23.8	19.4	14.4	10.5	9.8	11.8	15.5	19.9	22.7	24.9		
		MCWB	14.9	14.9	13.9	11.2	9.8	8.2	7.1	7.2	9.1	11.1	12.7	13.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	20.5	18.6	15.7	13.6	12.0	10.8	11.6	13.3	17.4	17.4	18.6		
		MCDB	25.4	26.9	25.1	20.9	15.4	12.8	11.7	15.5	19.2	24.0	26.3	25.7		
	2%	WB	18.6	18.8	17.2	14.5	12.5	10.5	9.4	10.1	11.8	14.6	16.1	17.0		
		MCDB	26.1	25.8	23.4	19.1	14.8	11.8	10.2	13.2	16.9	21.4	23.7	25.1		
	5%	WB	17.5	17.7	16.0	13.5	11.5	9.6	8.6	9.1	10.8	13.1	15.0	15.9		
		MCDB	25.7	24.8	22.8	18.4	13.8	10.8	9.4	11.4	15.3	19.3	21.7	24.0		
	10%	WB	16.3	16.5	14.9	12.6	10.7	8.9	7.9	8.2	9.9	11.9	13.7	14.7		
		MCDB	25.1	24.2	21.4	17.1	12.9	9.9	8.8	10.3	13.9	18.0	20.6	22.7		
(35) Mean Daily Temperature Range	MDBR		10.0	9.3	8.0	6.0	3.9	3.2	3.2	4.1	5.7	7.7	8.7	9.5		
	5% DB	MCDBR	13.0	11.7	10.4	8.3	5.5	4.6	4.5	6.7	9.2	11.4	12.0	13.1		
		MCWBR	5.0	4.9	4.2	3.7	3.1	3.2	3.1	3.7	4.3	5.1	5.0	5.4		
	5% WB	MCDBR	11.8	10.1	10.1	7.5	4.7	3.9	3.7	5.8	8.0	10.6	10.9	11.6		
		MCWBR	5.8	5.3	5.0	4.1	3.3	3.2	3.1	3.9	4.5	5.6	5.6	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.311	0.297	0.290	0.282	0.275	0.274	0.284	0.302	0.313	0.316	0.308		
	taud		2.585	2.606	2.643	2.643	2.648	2.653	2.648	2.602	2.541	2.511	2.536	2.586		
	Ebn at Noon		1025	1005	985	936	891	872	891	929	962	991	1014	1033		
	Edh at Noon		104	98	88	78	68	63	67	79	96	107	109	105		
(44) All-Sky Solar Radiation	RadAvg		7.79	6.83	5.55	4.02	2.75	2.27	2.46	3.38	4.63	6.01	7.10	7.65		
	RadStd		0.39	0.33	0.25	0.22	0.12	0.14	0.11	0.21	0.36	0.37	0.32	0.29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.90	N/A	N/A	+2.02	+0.64	N/A	-108	-275	+238	+56
(47) Regional (26 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-92	+157	+70

Nomenclature: See separate page